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Time: 20 to 25 minutes Total: 20 marks Tool: R (ggplot2 / base R). Work alone; your case file is different from your neighbour's. Write your answers and paste your plots and code.

Case File CD-C25: The Online store Case [20 marks]

Briefing: You are the data detective hired by the operations manager of an online store. The file holds 14 weekly records from the festive quarter, with seven variables. Some relationships can be trusted. Others are traps: a hidden driver, a curve that fools r, and a single wrong record. Find them.

R name	What it measures	R name	What it measures
festive_index	Festive-season index (0-100)	complaints	Customer complaints
ad_clicks	Advert clicks (hundreds)	staff_online	Support staff online
parcels	Parcels delivered (hundreds)	returns	Return requests
delay_dev	Delivery time minus promised time (hours)		

Evidence: run this code in R to create your case file (do not edit it).

```
set.seed(670721)
n <- 14
festive_index <- round(rnorm(n, 47.4, 17.0), 1)
ad_clicks <- round(68.9 + 22.7 * (0.92 * (festive_index - 47.4) / 17.0 + 0.38 * rnorm(n)), 1)
parcels <- round(51.7 + 17.0 * (0.88 * (festive_index - 47.4) / 17.0 + 0.47 * rnorm(n)), 1)
delay_dev <- round(seq(-27.3, 27.3, length.out = n)[order(runif(n))], 1)
complaints <- round(8.6 + 27.1 * (delay_dev / 27.3)^2 + rnorm(n, 0, 1.0), 0)
staff_online <- round(rnorm(n, 13, 4.5), 0)
returns <- round(22 + 9.1 * (0.20 * (staff_online - 13) / 4.5 + 0.98 * rnorm(n)), 0)
staff_online[12] <- 28; returns[12] <- 50   # row 12 was typed in by a trainee
case <- data.frame(festive_index, ad_clicks, parcels, delay_dev, complaints, staff_online, returns)
```

Tasks:

1. The matrix. Use cor(case) to get the 7 x 7 correlation matrix (round to 2 decimals). State n and how many distinct pairs of variables there are. Draw a correlogram of the matrix (ggplot2 geom_tile with geom_text, or corrplot). [4]
2. The strongest clues. Name the strongest positive pair and the strongest negative pair (off-diagonal). Give r for each and call it weak (below 0.3), moderate (0.3 to 0.7) or strong (above 0.7). [3]
3. The deceptive zero. Plot complaints against delay_dev with a straight line, then add a quadratic fit (formula = y ~ x + I(x^2)). Report r and the R-squared of the quadratic fit. In one sentence, say why r misses this relationship. [4]
4. The outlier alibi. Plot returns against staff_online. Find the row that looks wrong and give its row number. Report r for all rows and r without that row, and say what the single record did to r. [4]
5. The prime suspect. ad_clicks and parcels are strongly linked. Report r(ad_clicks, parcels), then r(festive_index, ad_clicks) and r(festive_index, parcels). Compute the partial correlation of ad_clicks and parcels controlling for festive_index: correlate resid(lm(ad_clicks ~ festive_index)) with resid(lm(parcels ~ festive_index)). Write a two-sentence verdict: does ad_clicks cause parcels? Name one study design that could test causation. [5]

Detective's rule: *correlation measures a straight-line link only, it can be inflated by one point, and it never proves cause.*

Code of Conduct: I certify that this work is my own. Signature: _____

22/10/26

Activity-07

Correlation Detective

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1) Correlation matrix

=> The dataset contains 114 weekly observations & 7 variables. The correlation matrix is obtained using cor(case), rounding values to two decimal places.

No. of distinct variable pairs: $\frac{7(7-1)}{2} = 21$

=> A Correlogram is used to visualize the strength and directio of Correlations. Positive correlations indicate that two variable tend to increase together, while negative correlation indicate an opposite relationship.

2) Strongest Correlation clues

Strongest positive pair:

=> Ad clicks and parcels. Both variables are strongly linked.

Strongest negative pair:

=> Identify the pair with the smallest off-diagonal correlation in the matrix.

=> Correlation strength is classified as weak when $|r| < 0.3$, modera when $0.3 \leq |r| \leq 0.7$, and strong when $|r| > 0.7$.

3) The deceptive zero

Plot customer complaints against delivery delay using a scatter plot. Add a straight-line fit and a quadratic fit.

=> The straight-line correlation may be close to zero even when the variables are related nonlinearly. The quadratic model can reveal a curved relationship that the straight line misses.

* Linear Correlation: r = output value.
* Quadratic model: R^2 = output value

Conclusion: A low linear correlation does not necessarily mean there is no relationship between two variables.

4) The outlier alibi

Plot returns against staff online. The observation that looks incorrect is row 12, because the staff-online value is -28., which is unusual for staff availability.

=> Calculate the correlation using all observations & again after removing row 12.

* Correlation with all rows: r = output value.
* Correlation without row 12: r = output value.

Conclusion: A single outlier can strongly influence the correlation coefficient and may create a misleading relationship.

5) The prime suspect

=> Calculate the correlation between ad clicks and parcels. Then calculate their correlations with festive index.

=> Use partial correlation to control for festive index.

Verdict: Ad clicks and parcels are strongly associated, but this does not prove that ad clicks cause more parcels to be delivered. Festive season demand may influence both variables.

=> A randomized controlled experiment, in which advertising exposure is randomly assigned to comparable groups, could help

establish whether advertising causes an increase in parcel deliveries

Code

```
Print(round(cor(case), 2))
f <- lm(complaints ~ delay_dev + I(delay_dev^2), case)
plot(case$delay_dev, case$complaints)
abline(lm(complaints ~ delay_dev, case))
cat("Linear r:", cor(case$delay_dev, case$complaints))
cat("\nAll rows:", cor(case$staff_online, case$returns))
cat("\nWithout row 12:",
cor(case$staff_online[-12], case$returns[-12]))
cat("\nclicks-parcels:", cor(case$ad_clicks, case$parcels))
cat("\nFestive-clicks:", cor(case$festive_index, case$ad_clicks))
cat("\nFestive-parcels:", cor(case$festive_index, case$parcels))
```

Output

